

Rian Touchent

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EDUCATION

Inria Paris - Sorbonne Université

Paris , France

Ph.D. in NLP - Team ALMAnaCH

2023 – 2026

- **Topic** : Information Extraction on French Electronic Health Reports
- **Advisers** : Laurent Romary and Eric de la Clergerie
- Published first paper about a new French biomedical model built using continual-pretraining called CamemBERT-bio
- Exploring RAG and LLM finetuning for biomedical information extraction

CentraleSupélec - Université Paris-Saclay

Gif-sur-Yvette , France

MSc in Artificial Intelligence

2021 – 2022

- Machine Learning and Deep Learning
- Reinforcement Learning
- Computer Vision and NLP
- Explainable AI

ECE Paris School of Engineering

Paris , France

Master's Degree in Engineering

2017 – 2022

- Maths (Linear Algebra, Probability, Calculus ...)
- C/C++, Java and C# programming
- Algorithms and Data Structures, Graph Theory
- Advanced Database and SQL

EXPERIENCE

Research Intern

June 2022 – November 2022

Inria - ALMAnaCH Research Team - Supervised by E. de la Clergerie ☞

Paris, France

- Research subject was adaptation of the CamemBERT model to the biomedical domain
- Collected several biomedical corpora using different techniques like filtering OSCAR or aggregating different sources
- Trained different versions of CamemBERT-bio using continual pre-training on GPU clusters
- Evaluated CamemBERT-bio on 3 french biomedical evaluation datasets on the NER task, improving F1-score by 1.9 on average
- Proposed to adapt the InstructionNER framework to the french biomedical domain using T5 for few-shots NER, beating SOTA by 1.51 on MEDLINE
- Introduced a new auxiliary task based on XML for InstructionNER improving F1-score by 2.05 on average on french biomedical data

Research Intern

May 2021 – August 2021

Inria - STARS Research Team - Supervised by A. Dantcheva ☞ and Y. Wang ☞

Sophia-Antipolis, France

- Research subject was Talking Face Generation which aims to synthesize a sequence of face images that correspond to given speech semantics.
- Evaluated and compared state-of-the-art audio to video generation techniques.
- Collected an Audio-Visual dataset using scraping methods.
- Proposed and trained a new architecture ☞ based on CNNs and GANs.

Deep Learning Intern

Winter 2020

Valtech_ - Recommendation letter ☞

Paris, France

- Trained an image segmentation model using deep learning techniques with Pytorch and Tensorflow/Keras (U-Net / M-RCNN)

- Initiated an image synthesis project using GANs (DCGAN, SPADE, StyleGAN)
- Hosted a presentation to explain deep learning and recent work to non-technical colleagues

Android developer for non-profit

Sep. 2018 – November 2018

Freelancing for LEHAÏM Handicap

Paris, France

- Development in Java & UI/UX of a donation app
- Developed an online in-app payment processing system with Stripe

PROJECTS

Insights extraction using NLP | *Spacy, PyTorch, Selenium*

End-of-studies project in collaboration with Renault

Sep. 2021 – April 2022

- Created a pipeline to automatically extract comments about Renault dealers on Google reviews using scraping methods
- Pre-processed and explored a newly created dataset using the pipeline
- Fine-tuned CamenBERT to affect each comment a step of the buying process
- Extracted insights about satisfaction levers from comments for each class

KASAR ☑ | Co-Founder, President

Student organization that helps students start their journey in AI

June 2019 – July 2021

- Managed a team of 7 vice-presidents
- Hosted 4 hackathons in Paris for beginners (≈ 30 participants per event)
- Mentored students personal projects
- Hosted a panel discussion with artificial intelligence experts

Klapp ☑ | Co-founder, CTO | *Flutter, React.js, Node.js, Python, Google Cloud Platform*

May 2017 - Present

- Working on front and back-end
- Published on iOS and Android gaining 200+ downloads
- Pitch in various project contests

Draft - Community-Driven News ☑ | *Flutter, Node.js, Python, Google Cloud Platform*

Sept. 2018 - 2020

- Published on iOS and Android gaining 1.5k+ downloads
- Top 3 of the day on Product Hunt

TECHNICAL SKILLS

Subject: Deep learning, NLP, Computer Vision, Mobile app development

Programming: Python, C/C++, SQL, JavaScript, Java, Dart, Lua

Frameworks: Flutter, React, Node.js

Dev Environment: Vim, Jupyter Notebook, VS Code, Google Cloud Platform, XCode, Android Studio, Ubuntu

Tools: Git, Conda, Virtualenv, SSH, Shell, LaTeX, wandb, Blender

Libraries: PyTorch, PyTorch Lightning, Keras

ACHIEVEMENTS

PAISS participant (2021) - AI summer school comprises lectures conducted by renowned experts like Yann Lecun, Cordelia SCHMID...

Merit Scholarship (2019) - Scholarship for being 1st/379 class rank at ECE Paris

Prologin semi-finalist (2018) - French National Computer Science Contest

ORGANIZATIONS

Humanitarian trip in Vietnam

January 2019

Helping an orphanage at common work and playing with children

Hue, Vietnam

Les petits frères des pauvres

2015-2017

Volunteer to support and help elderly people.

Paris, France

LANGUAGES

French: Native Proficiency

English: Full Professional Proficiency

Spanish: Elementary Proficiency

Arabic: Elementary Proficiency